



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC502 / 20EC503 - MICROPROCESSOR AND MICROCONTROLLER

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	With a neat diagram, explain the internal architecture of 8051 Microcontroller.	10	L*2	1	1
	b)	Write a program to add 10 two digit BCD numbers and store the higher byte result at 20H and 21H.	10	L3	1	1
2.	a)	Explain following instructions and also mention their addressing mode (i) XCH and XCHD (ii) SWAP A (iii) CJNE A,B,NEXT (iv) MOVC A, @A+DPTR	10	L2	1	1
	b)	Write as assembly language program to transfer 10 bytes of data from source location 40H-49H to destination location 45H-4EH with Overlapping. (with comments).	10	L3	1	1
3.	a)	Explain the internal organization of 8051 RAM with necessary diagrams.	10	L2	1	1
	b)	Write as assembly language program to sort the numbers in descending order (provide comments appropriately).	10	L3	1	1
		Unit – II				
4.	a)	Write an 8051 C program to send two different strings to the serial port. Assuming that SW is connected to pin p2.0, monitor its status and send the following message. SW=0; send "ELECTRONICS" SW=1; send "ELECTRICALS" Assume XTAL = 11.0592MHz, baud rate of 9600, 8-bit data, 1-stop bit.	10	L3	2	1
	b)	Write an 8051 C program to toggle all the bits of port P1 continuously with some delay in between. Use Timer 0, 16-bit mode to generate the delay.	10	L3	2	1
5.	a)	A door sensor is connected to the P1.1 pin, and a buzzer is connected to p1.7. Write an 8051 C program to monitor the door sensor, and when it opens, sound the buzzer. Buzzer can be sounded by sending a square wave of a few hundred Hz.	10	L3	3	1
	b)	Explain how interrupt priority can be changed using IP register. Also explain the default priorities assigned to interrupts in 8051 microcontroller.	10	L2	2	1
6.	a)	Explain two ways of connecting two slaves to a single master using SPI with the necessary block diagrams.	10	L2	2	1
	b)	Explain I2C protocol with the necessary timing diagram.	10	L2	2	1
		Unit – III				
7.	a)	Describe the features of the ARM architecture and draw the ARM visible registers.	10	L2	4	1
	b)	Explain 3 stage pipeline ARM organization with block diagram.	10	L2	4	1
8.	a)	Explain the function of following instructions with an example. i) MVN ii) MLA iii) BIC iv) RSB	10	L2	4	1
	b)	What will be the content in Register R0 after the execution of following instruction? Assume: R1 = 0x12345678 R2 = 0X01020304 R3 = C5000678 (i) MVN R0,R1, LSL#03 (ii) ADD R0,R1,R2, LSR,#02 (iii) MOV R0,R3,ASR #03	10	L2	5	1

BT* Bloom's Taxonomy, L* Level;

CO* Course Outcome; PO* Program Outcome
